

STARLAB Final Poster Rubric

Overview: For your final project, you will construct a scientific poster describing your research project and present the poster at the Fall Research Symposium at the end of the semester (alternatively, present at the Nebraska State Science Fair, a Lincoln High Symposium, et cetera, we are still working out the details).

Grading criteria for your poster: You will be graded on the rubric sections below for your poster. All list items are 'clickable' and will navigate you to the appropriate section of the document.

- [Hypothesis](#)
- [Research question](#)
- [Figures and Table Predictions](#)
- [Summary statement Predictions](#)
- [Justification of Predicted Results](#)
- [Methods and Design](#)
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Hypothesis: State the overall hypothesis for the study.

achieving	☐ hypothesis includes a 'because' statement that provides a rationale for making the prediction in the hypothesis based on previously published data or a proposed mechanism. ☐ can be supported or not supported by data presented in the poster.
progressing	☐ 'if' statement clearly recognizes and acknowledges what will be manipulated in the experiment (independent variable) ☐ 'then' statement clearly recognizes what the result of the experiment is and is measurable (dependent variable)
beginning	☐ is directly related to the experiments in the project ☐ statement contains a clear cause and effect

Feedback:

Research Question: Restate your hypothesis as a general question to help the reader understand what you are investigating in a big-picture sense.

achieving	☐ question is phrased so that it is answered by the prediction and justification. ☐ scope of question includes all of the data shown in the poster.
progressing	☐ includes clear independent variable/s (what is manipulated). ☐ includes clear dependent variable/s (what is measured).
beginning	☐ is within the scope of the project. ☐ is written as a question in the future tense ("Will...?")

Feedback:

Figure and Table Predictions: What do you expect your data to look like? Add a set of hand-drawn figures (usually one or two) that address the hypothesis. A table can replace a figure if it is more appropriate for addressing the hypothesis. Each figure or table should have its own caption.

achieving	☐ Captions do not include the word 'shows' ☐ Graphs have letters or stars over means to indicate statistical significance ☐ Captions do not include conclusions or statements summarizing statistical tests ☐ Captions indicate sample size as appropriate
progressing	☐ Figures are presented in a logical order and numbered in order ☐ Captions are found <u>below graphs and images</u> and <u>above tables</u> ☐ The descriptions in captions for figures and tables include the independent and dependent variables shown on the graph or table ☐ Each figure, table, or data image has a caption that starts with a label (e.g. " Figure 1... ").
beginning	☐ Figures or tables are not blurry and are clearly readable ☐ Each caption includes a basic description of what is shown ☐ A combination of at least two figures, tables, or images showing predicted experimental results are included

Feedback:

Summary statement predictions: Predicted summary statements should be added that state the main

conclusion of your figure/table predictions. Write one summary statement to accompany each figure or table prediction.

achieving	☐ Predicted summary statements adequately explain the experiment in question so that a reader can understand what occurred in the experiment. ☐ Predicted summary statements accurately interpret employed statistics and include appropriate values (p-values, correlation coefficients, etc.)
progressing	☐ Each summary statement highlights the data relevant to the hypothesis of the project. ☐ The conclusions presented in each summary statement are accurate with respect to the predicted data presented.
beginning	☐ Summary statements describe the main conclusion of the figure predictions. ☐ A summary statement is present for each figure prediction included in the poster. ☐ At the end of each statement, the figure prediction the statement is referring to is indicated in parenthesis.

Feedback:

Justification of Predicted Results: This is a statement that answers why you expect to get the predicted results you present based on previous published research. (Previous results show... This supports my research because...)

achieving	☐ justification is based on previous research ☐ the message is descriptive but to the point
progressing	☐ aligns with the data shown in the predicted results ☐ provides a justification with a biological basis
beginning	☐ a justification is attempted that is relevant to the research question

Feedback:

Methods and Design: In this section, you will provide a short summary the methods you plan to employ and your experimental setup along with a picture or diagram to illustrate the methods and design you intend to use.

achieving	☐ graphics or a flow-chart is used where appropriate ☐ includes appropriate detail to understand the predicted results ☐ does not include details that are unnecessary or unrelated to the predicted results
progressing	☐ written so a non-expert can understand ☐ methods are presented in a logical manner
beginning	☐ major steps for each method or procedure are included ☐ includes the methods that would be needed to make the predicted figures and tables

Feedback:

Introduction: In this section, you will provide background information about your subject (usually your dependent variable) and variable (usually your independent variable). Your introduction will start with the big-picture topic in your research question and will lead naturally to your hypothesis. Your background images will accompany your introduction.

achieving	☐ background images illustrate the broader subject the experiment is about and help provide context for the results ☐ flows naturally from a big-picture topic to a research question ☐ appropriately includes and cites at least two sources ☐ appropriately describes (and cites) the results of a study related to the experiment
progressing	☐ background images are appropriate to the study ☐ a knowledge gap is identified ☐ provides sufficient background information about the variables in the experiment ☐ appropriately includes and cites at least one source
beginning	☐ background images included ☐ starts with a statement about the broader topic the experiment is addressing ☐ information about the research topic included

Feedback:

Acknowledgments and References:

achieving	☐ all references cited in the poster are included and are formatted using a standard convention (e.g. APA) ☐ people listed in the acknowledgments have provided meaningful assistance to the project
progressing	☐ all references cited in the poster are included ☐ people who are listed in the acknowledgments are given specific credit for how they helped
beginning	☐ at least one reference used in the poster (usually introduction) is included ☐ anyone except for your partner/s and professor who helped with the experiment is listed

Feedback:

Poster formatting:

achieving	☐ there is spacing between sections, but no large areas of white space ☐ poster has a clear organization and flow
progressing	☐ poster is generally well organized ☐ all parts of the poster are readable in an appropriate font and size
beginning	☐ there are minimal spelling and/or grammatical errors ☐ template instructions and placeholders are removed

Feedback:

Video presentation:

achieving	☐ can be understood by a non-expert ☐ predicted conclusions are related back to the hypothesis ☐ provides background on why the hypothesis was explored
progressing	☐ provides at least one conclusion ☐ describes the research question or objective and hypothesis of the study ☐ describes the approach/es used to study the hypothesis
beginning	☐ describes the major predicted results of the experiment

	☐ is between 2 and 5 minutes ☐ is correctly linked to the poster using a non-expiring QR code
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Feedback: